

Occupied-Frame Validation for a U(1)-Symmetric Gaussian Transfer Circuit

Automated validation campaign

(Dated: August 22, 2026)

We validate a frame-native implementation of the canonical CPU Markov circuit against its rank-one covariance implementation at $N_x = N_y = 16$ for random pure and maximally mixed initial states. The first physics gate is a uniform $\alpha = 1$, domain-wall-free Chern insulator. The combined numerical and physics gate passed. Performance data, when present, are descriptive and are not part of implementation validity.

I. REPRESENTATION AND UPDATE RULES

For a pure physical Gaussian state, the occupied frame $V \in \mathbb{C}^{N \times r}$ obeys $V^\dagger V = I$ and $C = VV^\dagger$. For a mixed physical state we use a doubled pure frame $F = (A; B)$ with $C = BB^\dagger$; maximal mixing starts from $F = 2^{-1/2}(I; I)$. Given a normalized orbital e , gain uses $z = F^\dagger e$, $h = e - Fz$, and appends $h/\|h\|$. Loss builds a stable complex Householder reflector taking z to the last coefficient coordinate, applies its rank-one right action, and deletes the final column. Correct occupied and empty projections are chronological loss-gain and gain-loss words. Under perfect correction, mismatches reduce to a bare loss or gain.

The evolution kernel evaluates overlaps from the local $n_{\text{shell}} = 1$ support. It performs no covariance construction, pseudoinverse, or global QR. Covariance and doubled-projector construction occur only in the instrumented correctness observers and declared checkpoints; those costs are excluded from update-only benchmark timing.

II. CAMPAIGN DESIGN

The locked campaign uses $N = 512$ physical orbitals, 32 cycles, complex128 arithmetic, $\alpha_1 = \alpha_2 = 1$, no domain wall, and ten paired records per initialization. Ten single-sample worker processes are pinned to distinct physical cores on one NUMA node. Backend order is counterbalanced five/five, and workers synchronize before each timed backend phase.

III. CORRECTNESS

The gate passed. Maximum observed errors were:

maxmix: branch_probability_error=1.427e-11, choi_relative_error=2.547e-14, covariance_max_error=5.507e-12, covariance_relative_error=1.737e-12, frame_gram_residual=1.150e-14, global_entropy_error=3.174e-10, log_weight_error=1.933e-11, real_space_chern_error=1.017e-12, regional_entropy_error=1.249e-11.

random_pure: branch_probability_error=5.046e-12, covariance_max_error=8.062e-12, covariance_relative_error=1.512e-12, frame_gram_residual=1.121e-14, global_entropy_error=5.598e-11, log_weight_error=1.512e-11, real_space_chern_error=5.977e-13, regional_entropy_error=2.576e-11.

Any branch disagreement, nonfinite value, dense covariance fallback, missing shard, or rank inconsistency is a hard failure. Full covariance and doubled-Choi comparisons are diagnostic-only operations.

IV. UNIFORM-INSULATOR CONVERGENCE

The finite-size target projector gives $C_{\text{target}} = 0.999465$. Cycle-resolved disk-partition real-space Chern estimators are retained independently for all four initialization/backend lanes. Half-system entropy is retained for every lane, while global entropy density supplies the purification diagnostic for the maximally mixed initialization.

maxmix.covariance: $C_0 = 0.0000$, $C_{32} = 0.9999$, $S_{mglobal}(32)/N = 7.373e - 16$.

maxmix.frame: $C_0 = 0.0000$, $C_{32} = 0.9999$, $S_{mglobal}(32)/N = 0.000e + 00$.

random_pure.covariance: $C_0 = 0.0715$, $C_{32} = 0.9996$, $S_{mglobal}(32)/N = 1.700e - 15$.

random_pure.frame: $C_0 = 0.0715$, $C_{32} = 0.9996$, $S_{mglobal}(32)/N = 1.009e - 16$.

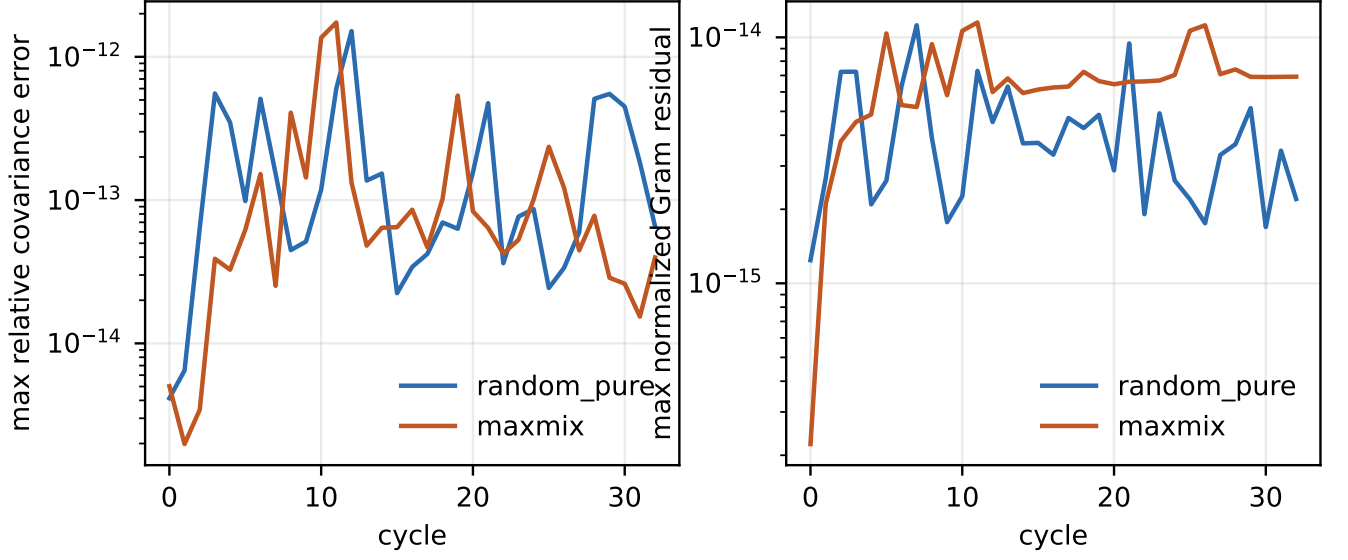


FIG. 1. Cycle-resolved covariance and Gram errors.

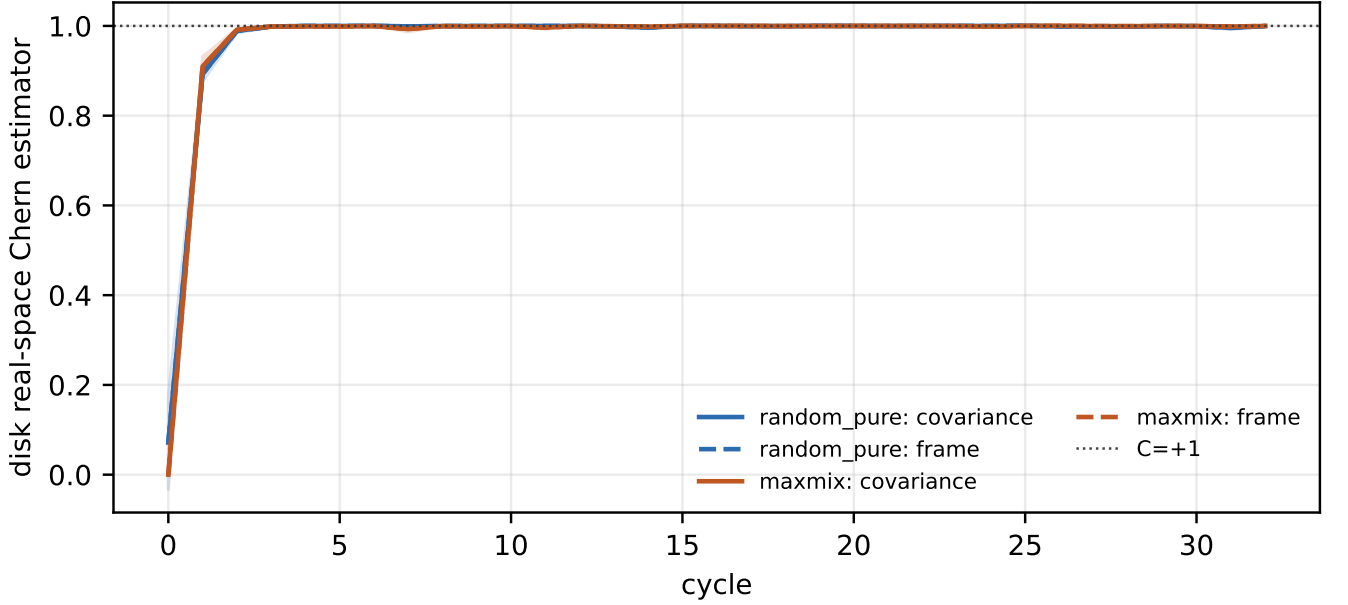


FIG. 2. Real-space Chern convergence for the covariance and frame replays of both initial-state families.

V. PERFORMANCE

For maxmix, the update-only covariance/frame ratio was 1.029 with paired-bootstrap 95% interval [1.014,1.043] (comparable); the end-to-end ratio was 1.026 [1.012,1.041] (comparable). For random_pure, the update-only covariance/frame ratio was 6.285 with paired-bootstrap 95% interval [5.972,6.648] (faster); the end-to-end ratio was 6.258 [5.953,6.612] (faster). Ratios greater than one favor the frame backend. Classification uses a predeclared $\pm 5\%$ practical-equivalence band and the paired-bootstrap interval, with each record as the independent unit.

VI. LIMITATIONS

This study covers a CPU-only 16×16 , $n_{\text{shell}} = 1$ uniform controller and 32 cycles. The disk Chern estimator is a finite-size convergence diagnostic; the calibrated target is therefore used instead of assuming an exactly quantized finite-lattice value. Purification frames retain doubled-space costs and may show no speed improvement. Observer Chern contractions, eigendecompositions/SVDs, and diagnostic reconstructions are reported separately from dynamics.